



12.3.5 Wrap-Up: Ticket Out the Door

A recipe for sparkling grape juice calls for $1\frac{1}{2}$ quarts of sparkling water and $\frac{3}{4}$ quart of grape juice.

1. How much sparkling water should be mixed with 9 quarts of grape juice?
2. How much grape juice should be mixed with $\frac{15}{4}$ quarts of sparkling water?

Unit 12, Lesson 4: Discounts and Markups



Benchmark

MA.7.AR.3.1 Apply previous understanding of percentages and ratios to solve multi-step, real-world percent problems.



Learning Target

- ☐ I can solve problems involving discounts and markups.



12.4.1 Warm-Up: Civil Engineer

Alishia Ballard works as a civil engineer for San Francisco Public Works. Civil engineers work on the plans to create structures such as bridges, large buildings, etc. They also analyze whether or not structures are safe.

1. How do you think civil engineers use math in their tasks?

Alishia enjoys working at Public Works because it's a way to give back to her community. Every structure plan she works on or structure she inspects is all for the people around her.

2. What is something you have done that gives back to your community, your school, or your peers?



12.4.2 Guided Instruction: Discounts and Markup

Percentages and ratios are often used to express or solve problems involving changes in the price of goods and services.

1. The price of a shirt is \$18.50. Natalie has a coupon that reads "discount $\frac{1}{2}$ off."
 - a. With the coupon, will Natalie pay more or pay less than \$18.50 for the shirt?
 - b. Discuss with your partner what " $\frac{1}{2}$ off" means. Summarize your discussion.

A **discount** is a reduction of a price for goods or services.

- c. By what percentage will the coupon reduce the price?

d. What is the discount amount, in dollars, taken off the shirt price?

e. What is the discounted price, in dollars, of the shirt after the coupon is applied?

2. Cassidy, a young inventor of Baby Toon, appeared on Shark Tank in 2019. Cassidy shared that it costs her \$6.60 to produce one of her baby spoons and that she sells them to customers marked up to \$15.00.

a. Does a markup increase or decrease the original price?

b. Discuss with your partner what you think markup means. Summarize the discussion by writing a definition for markup.

c. Complete the statements.

A markup is applied to goods or services so that the

seller is able to make a

- ☐ loss.
- ☐ profit.

The markup

- ☐ increases
- ☐ decreases

the price from the cost to the seller

and is usually determined using percentages.

d. If Cassidy placed a 100% markup on her spoons that only cost \$6.60 to produce, what would the price of a spoon be for customers?

e. Discuss with your partner how you determined a 100% markup of \$6.60. After your discussion, make any revisions to part b, if necessary.

- f. Jovi and Malik used two different methods to find the price of the spoon if the markup used is 61%. Part of their work is shown.

Jovi's Work	Malik's Work
$\frac{m}{6.60} = \frac{61}{100}$ $100 \cdot \frac{m}{6.60} = \frac{61}{100} \cdot 100 \cdot 6.60$ $100m = 402.6$ $m = 4.026$	$0.61(6.60) = m$ $4.026 = m$

Complete the statements.

Jovi used a percent
☐ equation
☐ proportion
 to find the value

of m while Malik used a percent
☐ equation.
☐ proportion.
 Both

methods resulted in the same value of m , which

represents the
☐ dollar amount of the markup.
☐ price of the spoon after markup.

At a 61% markup, the spoon would sell for
☐ \$4.02.
☐ \$4.03.
☐ \$10.62.
☐ \$10.63.

- g. Jovi and Malik both found the price of the spoon after a 61% markup by calculating the amount of the markup and adding it to the cost of the spoon. What percentage could they have used if they wanted to calculate the price after markup in one step?



12.4.3 Activity: Math Menu

With your partner, choose your meal from the math menu shown. Select one appetizer, one main course, two side dishes, and one dessert. Put a * next to your selections. Then, solve each problem selected.

Appetizers

- A1. A book has been discounted by $\frac{3}{10}$. The original price of the book is \$34.99. What is the amount being discounted from the book?
- A2. Westside Skate Shop is having a storewide $\frac{1}{5}$ off sale. Find the amount saved on a \$66 skateboard.
- A3. Jennea bought a scratched iPhone because it was discounted $\frac{9}{20}$ off the original price of \$795. What was the reduced price, in dollars?

Main Courses

- M1. Ramin's Discount Automobiles buys a used car for \$3,000. They mark it up 65% before selling it to a customer. What was the sale price of the car?
- M2. Elliott sells grab-and-go lunches. He spends \$2.70 per lunch on ingredients and then adds a 110% markup to sell. What is the selling price of one sandwich?
- M3. It costs JS Surfboards \$225.00 in labor to make one Model S surfboard. The company marks up the surfboard 51% to sell. What amount, in dollars, are Model S surfboards sold for?

Side Dishes

- S1. A concert ticket is on sale for \$129. The sign says 14% off the original price. What was the original price of the ticket?
- S2. A scientific calculator is 15% off. The sale price of the calculator is \$13.99. What is the original cost of the calculator?
- S3. The price of an apple is \$1.25. If you get a 10% discount, how much do you have to pay?
- S4. Teena bought a bicycle for \$528 after getting a discount of 12% taken off its normal price. Find the normal price of the bicycle.

Desserts

- D1. A store sells notebooks for \$1.60 after a $\frac{3}{10}$ markup. How much is the markup?
- D2. Burgermonster sells its burgers for \$1.84. To determine the selling price, the restaurant added a markup of $\frac{23}{10}$ to the cost of the ingredients used to make a single burger. What is the cost Burgermonster pays for ingredients?

Appetizer Selection Work and Answer**Main Course Selection Work and Answer****Side Dish Selection Work and Answers****Dessert Selection Work and Answers**



12.4.4 Wrap-Up: Reflection

1. What was the **muddiest** (most difficult or confusing) point in today's lesson on discounts and markup?



Unit 12, Lesson 5: Determining Tax, Tip, or Fees Given a Percentage



Benchmark

MA.7.AR.3.1 Apply previous understanding of percentages and ratios to solve multi-step, real-world percent problems.



Learning Targets

- ☐ I can determine the amount of tax, tip, or fee given a percentage.
- ☐ I can determine the percentage of a discount or markup.